

VIJAYAWADA INTL, India

WMO: 431810

Lat: 16.530N

Lon: 80.797E

Elev: 82

StdP: 14.65

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	65.2	66.9	55.7	66.1	77.5	57.9	71.8	77.7	10.0	84.3	9.6	84.1	0.3	10	0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	20.0	106.0	80.2	103.6	80.4	100.7	80.2	84.1	95.3	83.4	93.6	82.7	92.0	6.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.9	166.6	88.0	81.0	161.6	87.4	80.6	159.6	87.1	48.4	95.6	47.5	93.4	46.8	92.5	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.9	11.3	9.7	DB	60.9	112.0	1.9	2.6	59.5	113.8	58.4	115.4	57.3	116.9	55.9	118.8
			WB	57.9	85.4	1.4	1.3	56.9	86.4	56.1	87.1	55.3	87.8	54.3	88.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.2	76.9	79.8	85.2	90.1	93.2	89.4	85.2	84.6	84.4	83.6	80.4	77.8
	DBStd	5.67	1.98	2.31	2.42	2.74	3.78	5.40	3.77	2.91	2.29	2.51	2.43	2.35
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12495	834	835	1090	1204	1338	1182	1091	1074	1032	1042	912	861
	CDD65	7020	369	415	625	754	873	732	626	609	582	577	462	396
	CDH74	80510	2366	3292	6837	10140	12850	10267	7702	7203	6779	6314	4017	2743
	CDH80	39785	695	1388	3362	5927	8425	6100	3536	3025	2708	2453	1314	852
(14) Wind	WSAvg	3.3	2.1	2.1	2.4	3.3	4.0	4.7	4.8	4.4	2.8	2.4	2.7	3.1
(15) Precipitation	PrecAvg	48.7	0.3	0.4	0.5	0.8	2.7	6.0	9.1	9.0	8.2	8.3	2.7	0.8
	PrecMax	73.9	1.5	2.0	5.2	4.3	15.4	11.7	13.8	15.3	16.6	15.6	8.1	4.3
	PrecMin	35.3	0.0	0.0	0.0	0.1	0.5	2.4	5.4	5.0	2.7	0.9	0.0	0.0
	PrecStd	8.5	0.4	0.6	1.0	0.8	3.1	2.6	2.2	2.1	3.2	4.2	2.4	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.0	95.1	100.4	105.6	111.3	108.0	100.2	96.8	95.3	95.1	91.5	89.8
		MCWB	71.7	73.8	77.7	80.4	78.0	80.5	79.0	79.7	81.1	79.1	76.7	73.0
	2%	DB	86.8	92.9	97.2	102.5	107.6	104.6	96.7	94.7	93.5	93.2	89.8	87.9
		MCWB	71.5	73.7	77.3	80.6	79.4	80.2	79.2	79.9	80.6	79.2	76.1	73.0
	5%	DB	85.8	90.0	95.3	100.6	104.4	101.9	94.4	92.9	91.8	91.5	88.2	86.3
		MCWB	71.4	73.4	77.1	80.2	80.5	79.8	79.4	80.2	80.6	78.7	75.8	72.7
	10%	DB	84.2	87.9	93.5	98.7	102.0	98.5	91.7	90.7	90.0	89.8	86.7	84.8
		MCWB	71.1	72.8	76.8	79.7	80.7	79.9	79.5	79.8	80.4	78.7	75.5	72.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.4	78.7	81.3	84.2	85.4	84.8	83.2	83.4	83.7	82.8	80.1	77.2
		MCDB	83.4	90.3	93.2	97.3	99.7	96.9	90.9	91.2	91.1	89.9	86.0	84.8
	2%	WB	75.3	77.0	80.3	83.0	84.4	83.6	82.1	82.4	82.6	81.8	79.2	76.1
		MCDB	81.4	86.9	91.7	94.5	96.4	94.4	89.4	89.6	89.5	88.5	85.3	83.3
	5%	WB	74.1	75.8	79.4	82.2	83.6	82.6	81.4	81.6	81.8	81.0	78.3	75.2
		MCDB	80.8	84.6	90.1	92.9	94.6	91.7	88.6	88.3	88.3	87.3	84.3	82.2
	10%	WB	72.8	74.7	78.5	81.5	82.8	81.8	80.6	80.8	81.1	80.3	77.4	74.2
		MCDB	80.2	83.0	88.4	91.9	93.2	90.7	87.7	87.0	87.2	86.3	82.9	81.0
(35) Mean Daily Temperature Range	5% DB	MDBR	17.8	19.9	20.4	20.2	20.0	15.3	11.6	11.5	11.6	13.0	14.5	16.8
		MCDBR	18.3	21.9	21.4	21.5	22.3	20.2	15.6	14.3	14.0	15.3	15.8	17.7
		MCWBR	5.9	6.3	5.6	4.9	5.5	5.0	4.3	4.7	4.9	4.9	5.4	5.8
	5% WB	MCDBR	15.9	19.4	19.9	20.8	20.4	16.8	12.8	12.6	12.6	13.0	12.8	15.1
		MCWBR	6.4	7.2	5.7	5.4	6.0	5.7	5.2	5.3	5.4	5.0	5.0	5.9
(40) Clear-Sky Solar Irradiance	taub		0.521	0.513	0.571	0.587	0.653	0.625	0.575	0.560	0.556	0.567	0.536	0.551
	taud		1.949	1.995	1.879	1.885	1.750	1.822	1.958	2.001	2.006	1.959	1.978	1.902
	Ebn at Noon		242	253	242	238	220	224	236	242	243	235	236	229
	Edh at Noon		57	56	65	65	74	68	59	57	57	58	55	58
(44) All-Sky Solar Radiation	RadAvg		1491	1748	1916	2067	1960	1559	1389	1432	1498	1486	1441	1374
	RadStd		59	74	73	76	64	128	159	114	117	123	95	87

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.54	+0.58	+0.86	N/A	+0.36	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page